

spaceThink — Condensed Brief

An EXHYTE agent for autonomous scientific reasoning over deep-space telemetry.
Prepared 2026-07-20. This is the short read. Every claim here is backed by a fuller document listed in the Reference Index at the end — request any of them for detail.

Executive summary

We took your research document, ran a literature review across four fronts (90+ sources, every citation verified), and turned it into a specification, a build plan, and a market read. The short version: **your thesis holds**. The gap you identified — telemetry AI that detects but never explains, and never closes the loop from anomaly to hypothesis to test — is real in both the current research and the commercial market. Nobody sells the closed loop yet. The architecture you proposed is sound, and the pieces to build it now exist off the shelf.

Two things came out of the review that you should see. First, a few technical assumptions in the paper need updating (the standard NASA benchmarks are now considered broken; feeding raw telemetry to an LLM loses to simple statistics; the winning pattern is a hybrid). Second, and more important, there's a **strategic fork** we need to decide together: how far to lean toward autonomous *scientific discovery* (the ambitious vision) versus spacecraft *health diagnosis* (the fundable near-term product). Both run on the same engine. That decision, not any technical detail, is the thing this brief is really asking you to weigh in on.

What the review confirmed

Three findings, each independently verified:

The framework is real and citable. EXHYTE is a December 2025 survey (Hasib et al.) that formalizes discovery as Explore → Hypothesize → Test with feedback loops. Its own assessment: exploration is a mature research area, but hypothesis *prioritization* and the entire *testing* stage are the least-developed parts of the field. Our product lives in exactly those two weak spots — which is both the risk and the opportunity.

The architecture has been proven next door. A November 2025 paper (AURA) does LLM + digital-twin + human-in-the-loop diagnosis for underwater robotics — structurally identical to what we're proposing. A 2026 paper does simulation-based

fault diagnosis on an engineered system with an 82× speedup over the old method. The feasibility risk is largely retired. What’s left is a race to the spacecraft vertical before someone else runs it.

The market has the same hole as the literature. Funded companies either detect anomalies (Intella, Constellation Space), automate operator workflow (Quindar, Leanspace), or sell simulation twins (Sedaro, Antaris). None generate a mechanistic root cause and *verify it in simulation* before showing it to a human. That closed loop is unclaimed. Meanwhile the demand driver is strong: the active satellite count has tripled since 2019 and could hit 50–100k by 2030, manual operations break down past roughly 5–10 satellites per team, and a single 2023 anomaly (ViaSat-3) produced a \$420M insurance claim.

What the review corrected in the original paper

You wrote the paper before some of this evidence existed. Four updates:

- **The NASA SMAP/MSL benchmarks are no longer credible as headline results.** The field now considers them broken — trivial anomalies, mislabeled ground truth, and an evaluation trick (point-adjustment) under which even random scores look state-of-the-art. The real benchmarks are ESA-ADB (2024) and OPS-SAT-AD (2025), both real ESA telemetry with honest metrics.
- **Don’t feed raw numeric telemetry to an LLM.** Multiple 2025–2026 studies show LLMs lose to simple statistical detectors on raw time series. The winning pattern is hybrid: a lightweight detector owns the numbers, the LLM owns the *mechanism and explanation* over pre-digested features. An industrial version of this cut diagnosis time from 7.4 hours to 1.2 and raised accuracy sharply.
- **The Gramian-Angular-Field image encoding you cited underperforms** on ESA-ADB (it was the worst of three approaches). Dropped from the plan.
- **The best detector is surprisingly old.** A channel-wise LSTM forecaster with dynamic thresholding (the 2018 “Telemanom” lineage) still leads the ESA benchmark; transformers and time-series “foundation models” have not beaten it under honest evaluation. So the first thing we build is well-understood, not exotic.

None of this breaks the thesis. It sharpens it — and, usefully, “we evaluate honestly in a field full of inflated numbers” becomes a selling point.

The strategic fork (the decision we need)

The same engine — explore telemetry, hypothesize a mechanism, test it against a digital twin, refine — can be pointed at two different targets. What differs is *what the hypotheses are about*, and that changes who the customer is, what data we need, and how we prove it works. There are three defensible directions:

	A — Science-first	B — Mission + wedge	C — Ops-first
Aims at	New physical discoveries	Both, sequenced	Fault diagnosis
First customer	Science teams, agencies	Operators now → science later	Constellation operators
Demo	Autonomous <i>rediscovery</i> of a known phenomenon	Ops triage + one rediscovery	Detect → hypothesize → twin-verify
Revenue soon	Grants / mission-funded	Commercial SaaS funds the mission	Commercial SaaS
Vision fidelity	Highest	High	Low
Main risk	Hard to evaluate “a discovery”	Staying focused on both	It isn’t the vision

Direction A is closest to what you originally described — satellites and probes reasoning their way to genuinely new science. It’s the most inspiring and the most differentiated, but its customers are agencies, its revenue is grant-shaped, and “what counts as a discovery” is genuinely hard to measure. **Direction C** is a verified anomaly-diagnosis product for satellite operators — the clearest customer and revenue story, but it isn’t the company you set out to build, and it’s the version our first draft accidentally drifted into. **Direction B** is the diagonal: commercialize diagnosis first because it has customers and clean data, while the *stated mission* stays autonomous science, and the ops revenue funds the moonshot and buys the operator relationships the science layer will later need.

There’s also a second, independent axis: whether the agent only analyzes on the ground (offline) or eventually reasons and *acts* onboard under communication delay. That onboard autonomy — the “robot doing its own experiments” image — is a later-stage move in all three directions; the first version stays offline for safety and tractability regardless.

Our recommendation, if we don't want to choose in the dark, is **B**: it keeps the vision as the actual point of the company while giving us a fundable wedge that's already worked out in detail. But this is your call as much as mine — I've written all three up in full so we can compare them honestly rather than back into one.

The plan, in brief

Assuming we start from the diagnosis wedge (Direction C, which is also Phase 1 of B):

Stack, decided. Datasets: OPS-SAT-AD to iterate, then ESA-ADB as the benchmark we publish against. Simulator/twin: Basilisk (open, Python-native, models reaction-wheel and power faults, with published precedent for exactly this use). Detector: the Telemanom-lineage forecaster, conditioned on telecommands from day one (commanded maneuvers are the #1 source of false alarms). Hypothesis scoring: neural simulation-based inference (the `sbi` package) to score fault hypotheses against the twin. First subsystem: attitude control / reaction wheels.

The MVP demo. Stream a real ESA anomaly; the agent detects it, generates three ranked mechanistic hypotheses, runs each in the twin, and shows one reproducing the real telemetry signature — detection to verified root cause in under five minutes, next to the days a human review board takes. A second demo attacks the “muted alarm” pain: triage a day of noisy telemetry, auto-explain the benign 98%, escalate the two real faults with verified causes.

How we build it. Six phases (foundation → exploration/detection → twin + fault library → simulation-based scoring → hypothesis layer → planner + dashboard), each shipping a testable increment behind a quality gate. Roughly 12 weeks to a demo-ready loop with two people.

How we split it. The architecture divides cleanly into two halves along its deepest seam: one of us owns **data & physics** (ingest, detection, the Basilisk twin, the fault library), the other owns **inference & agent** (the simulation-based scoring, the LLM hypothesis roles, the planner). We freeze the shared contracts in week one, then each build behind the other's stub so neither of us ever blocks. It scales to 3–4 people by splitting the physics from the data and giving the agent/product layer its own owner. Full week-by-week schedule is in the work-split document.

Governance. The repo is already scaffolded and runs under an automated quality gate (build, lint, types, tests, security) that has to pass before anything is called done. The domain model and its tests are written; the gate is green.

What we need from you

1. **Weigh in on the fork** — A, B, or C (my lean is B). This reshapes the goals, the demo, and the customer, so it's worth deciding before we build much more.
2. **Sanity-check the corrections** to the paper's technical assumptions — you know this literature; tell me if any of the four updates look wrong.
3. **Ratify or push back on the wedge** (diagnosis-first) and the reaction-wheel starting subsystem.

Everything else is written down and ready to expand on request.

Reference index (request any for the full version)

The detailed documents behind this brief. All are in the project repo; each is also available as its own PDF.

- **Strategic Directions** — the three directions above, in full, with the two-axis map, per-direction customers/data/evaluation/risks, and decision criteria.
- **Product Specification** — requirements as user stories with acceptance criteria, measurable quality bars, the module architecture, and every scope decision with its rationale.
- **Implementation Plan** — the six build phases with exit criteria and the dependency graph.
- **Parallel Work Split** — the 2-, 3-, and 4-person plans, interface contracts, and the week-by-week parallel schedule.
- **Literature Review (Synthesis)** — how the four deep-dives confirm and correct the original paper, plus the finalized technical decisions.
- **Research: Telemetry Anomaly Detection** — benchmarks, SOTA methods, dataset practicalities (22 sources).
- **Research: LLM Scientific Discovery** — the EXHYTE framework, AI-scientist systems, what works and what fails (19 sources).
- **Research: Digital Twins & Simulation-Based Inference** — simulators, scoring methods, fault libraries (25+ sources).

- **Research: Market Landscape** — competitors, market size, customers, YC positioning, investor objections (24 sources).

The full 64-page dossier stitches all of these together in one file, if you ever want the whole thing.